# Investigating the Capability of RNO-G Stations through Coincidental Events and Signal Characterization


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## Abstract


The Radio Neutrino Observatory in Greenland (RNO-G) contains 10 operational stations that detect radio frequency (RF) signals from neutrino showers. Deep below the ice, these stations can trigger hundreds of times per minute due to RF signals from planes, snowmobiles, and other events. We present an investigation of detected events across multiple stations and findings about how signal properties can be characterized in ice. Our investigation of coincidental signals showed consistent results with what was expected for calibration runs. Our signal characterization results show that simulation matches calibration data relatively well for both amplitude ratio comparisons and time delay calculations in refracted and direct RF signals. Together, these topics provide clarity about possible neutrino events and how neutrino RF signals move and respond in ice.


## Introduction

- The Radio Neutrino Observatory in Greenland (RNO-G) has 10 operational stations, each with 24 channels to detect radio frequency (RF) signals
- As neutrinos go through ice, they interact and create secondary particles which result in RF signals to reach the station antennas

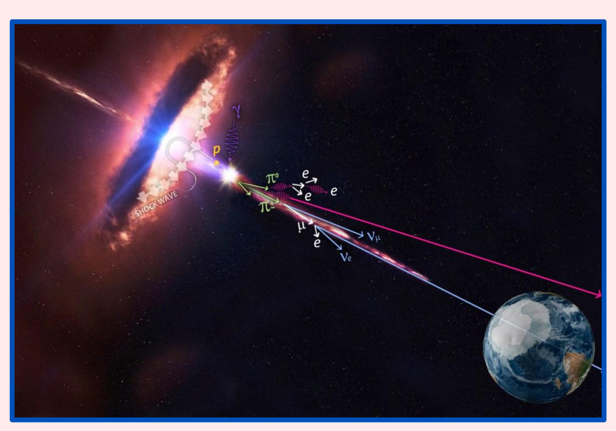

Fig. 1. Visual showing high energy events releasing neutrinos to Earth [4]

- Airplanes, snowmobiles, cell phones, and instrumental errors can cause stations to trigger
- 3 different types of antennas at each station: Surface LPDA's, Horizontally polarized (Hpol), and Vertically polarized (Vpol)
- Each antenna type responds differently to signals

Hpol antenna | Vpol antenna | LPDA

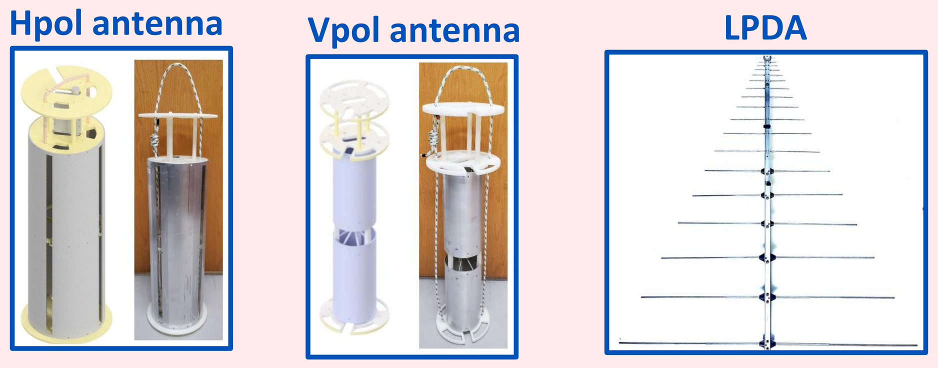

Fig. 2. Types of RNO-G station antennas [1]

- Neutrinos are typically not viewed at multiple stations, meaning coincidental events indicate a larger-scale phenomenon by non-astroparticle interactions
- Coincidence analysis can be crucial to validating neutrino candidate events
- Direct and refracted/reflected signals show different amplitude ratios and time delays at various signal origin depths within ice
- RNO-G collaboration has not done much work with station coincidences and direct/refracted signal analysis, prompting our investigation of such topics

## Coincidental Events

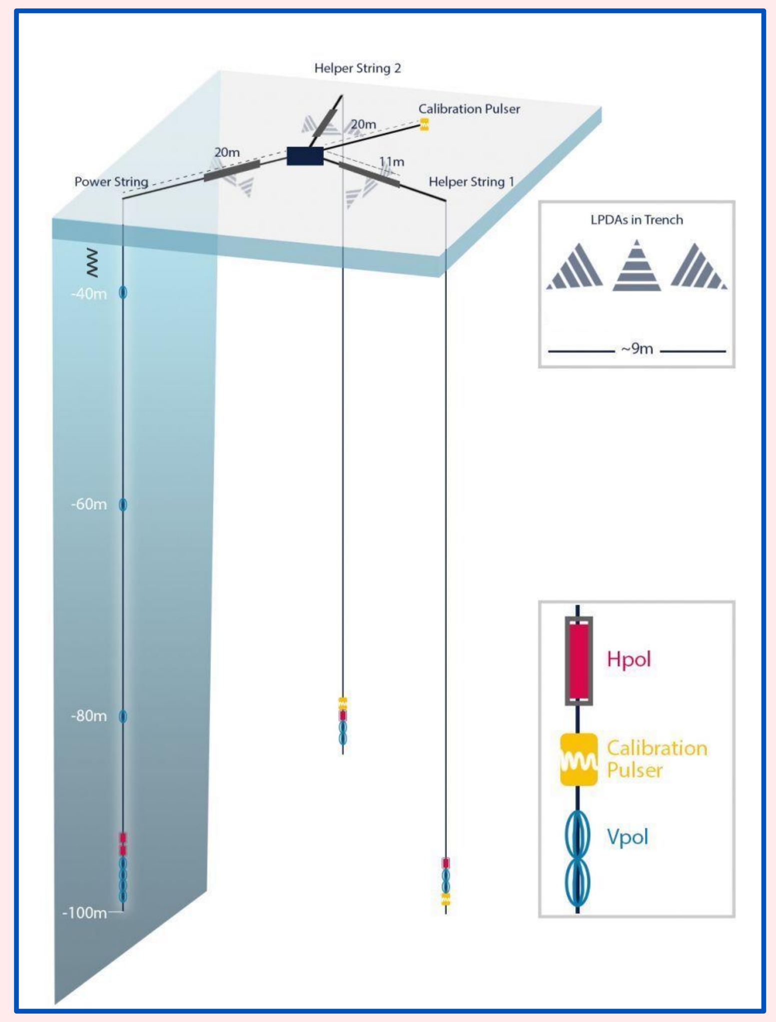


Fig. 3. Diagram of RNO-G station map, containing types of instruments and station depth [5]

- Each RNO-G station is created to look the same and are spread apart by 1.5 km
- We compared triggered events across all stations during 2025 recorded by RNO-G
- Over 300 returned coincidences within +/-30 microsecond frame

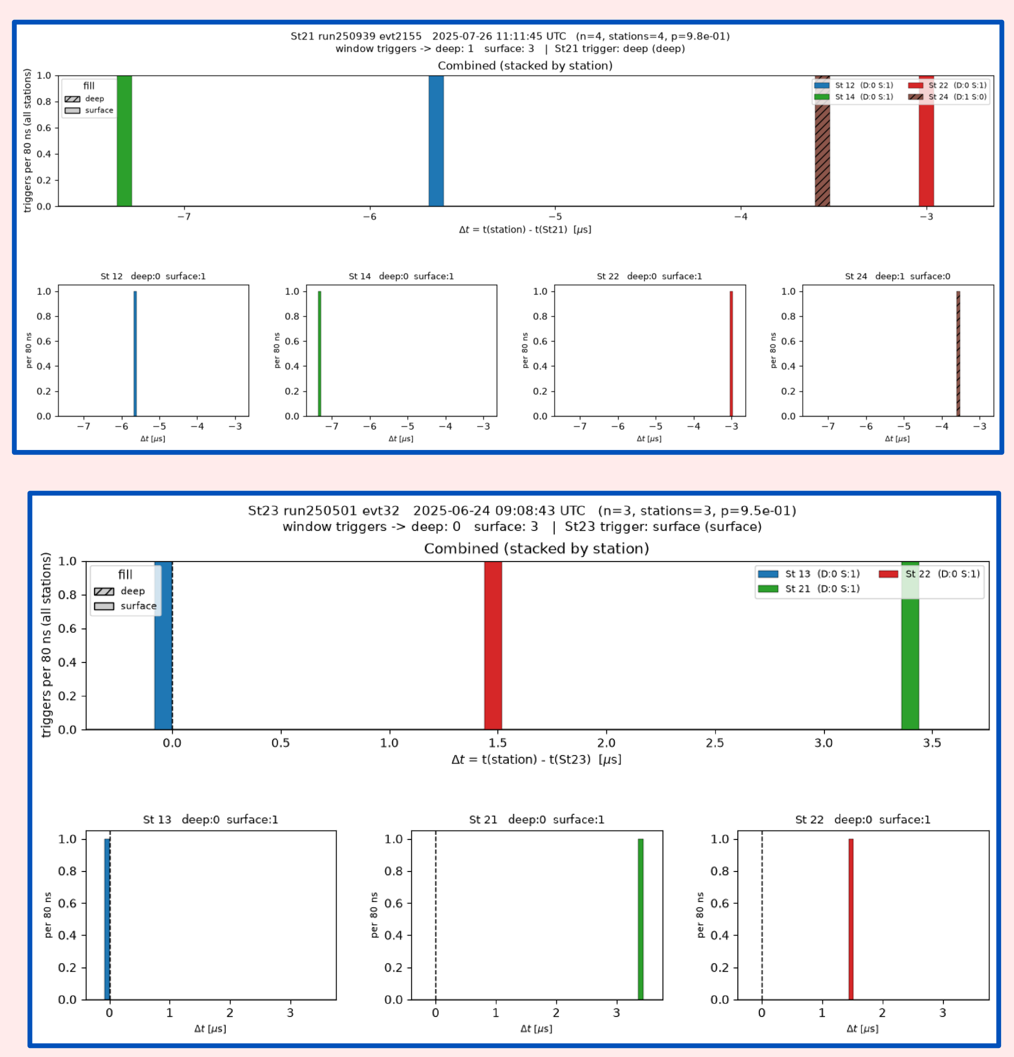

Fig. 4. Histograms produced showing the coincidental triggers of one event at a reference station, whole frame is 60 microseconds

- These histograms show which stations happen to trigger within +/- 30 microseconds of an event at a reference station
- Most neutrino events will not appear in more than one station
- Histogram analysis seems to be working well
- Saw an increase of triggered events at all stations when planes were documented overhead

## Signal Characterization

- RNO-G station geometry allows for a combination of direct, refracted, and reflected signals
- We chose station 21 calibration data to understand how pulser signals are transformed and delayed through ice
- Station 21 only has refracted and direct signals (no reflected)

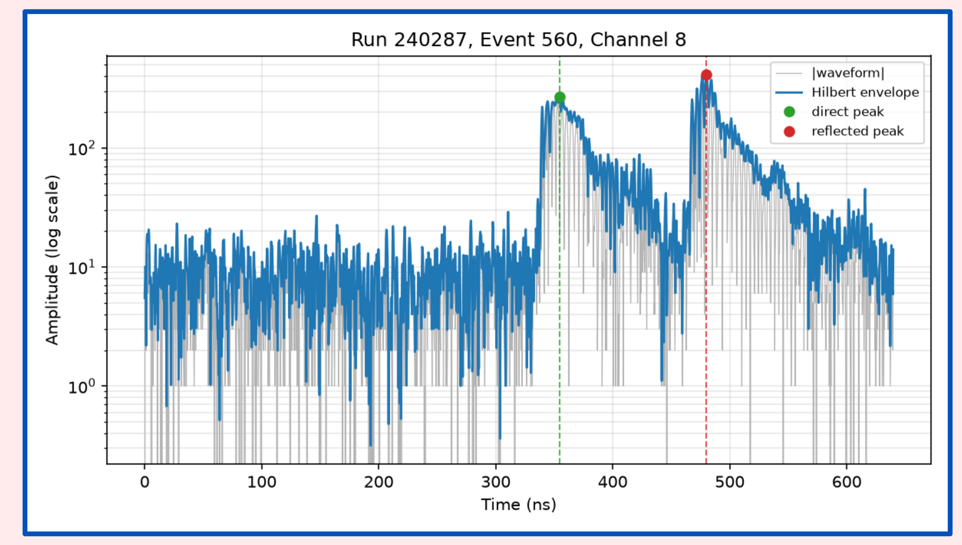


Fig. 5. Image from peak-finding algorithm to pick out direct and refracted calibration signals

- Time delays and amplitude ratio between refracted and direct were investigated
- These parameters are dependent on depth within the ice
- Real data was compared with NuRadioMC simulation to see the expected and resulting time delays and amplitude ratios

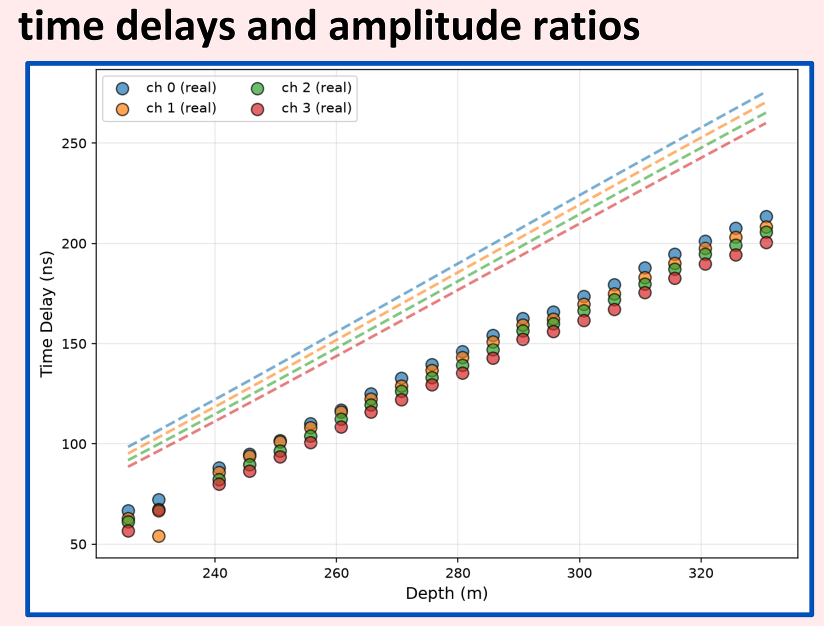


Fig. 6. Expected and resulting time delay between station 21 refracted and direct signals of same event (8-1-2024)

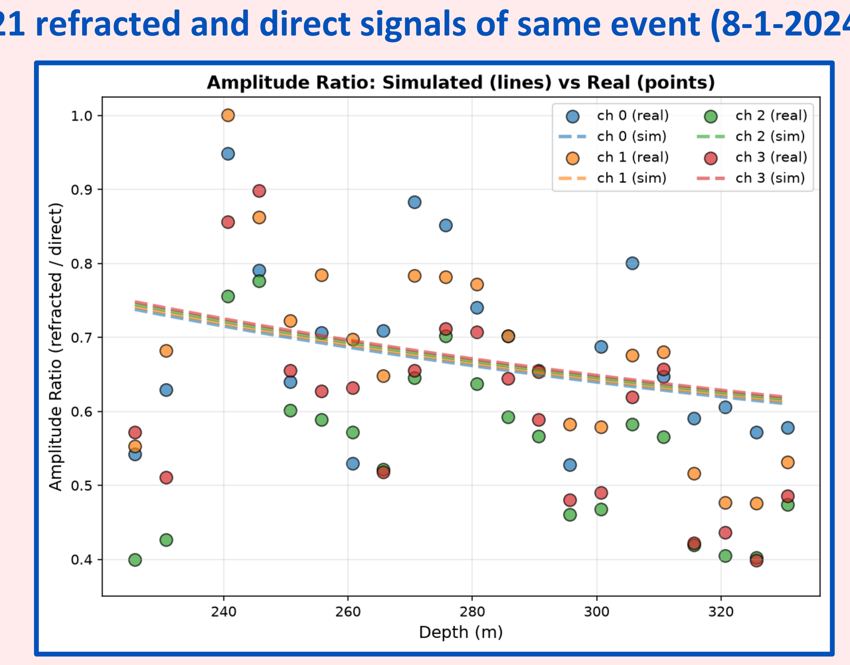


Fig. 7. Expected and resulting amplitude ratio between refracted and direct signals of same event (8-1-2024)

- Both figures show comparable values between simulated and real data
- Current model for ice might not exactly capture amplitude ratios properly

## Conclusions

### Coincidental Events

- Coincidence analysis scripts are working as expected, finding a rise in triggers across all stations when planes were overhead
- Coincidental analysis can eventually be used as another test to indicate if a neutrino candidate is valid or not

### Signal Characterization

- Real and Simulated data shows similar trends, but not exact accuracy to one another
- As the depth decreases, the refracted amplitudes get smaller compared to the direct amplitudes
- Time delay between refracted and direct gets larger with depth

## Future Work

- Create pipeline to hand-select direct and refracted signals for lower SNR channels
- Find better model for amplitude ratios
- Compare different Vpol heights within station for amplitude ratio and time delay
- Conduct coincidence analysis on all 2025 data and analyze the statistics

## References


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